Guest Review Intelligence & Action Management System

A Reputation-Protection Platform for The Kingsbury, Colombo

Final Module Report

Turning thousands of scattered guest reviews into a prioritised, department-owned action list, before problems define the brand.

Module	______________________
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A working proof-of-concept built on live analysis of real Kingsbury reviews crawled from Google, TripAdvisor and Booking.com, augmented with engineered incident fixtures. Hotel financial figures are illustrative and explicitly flagged.

Abstract

A five-star hotel's reputation is no longer written only by its front desk. Increasingly it is mediated by algorithms, such as search rankings and conversational AI assistants, that read guest reviews and distil them into a one-line verdict. When a recurring complaint is repeated by enough guests it becomes a permanent caveat in how the property is described on every query. The Kingsbury, Colombo, a five-star hotel rated 4.0/5 and ranked #32 of 103 hotels in Colombo on TripAdvisor, illustrates the problem. Thousands of reviews are spread across multiple platforms, far beyond what any operations team can read, cross-reference and act on consistently. This report presents a working proof-of-concept that runs the same mechanism that judges the hotel, but *for* the hotel. A multi-platform ingestion layer feeds a natural-language analysis pipeline that scores every review for sentiment and a 0–100 reputation-risk value. A three-pass large-language-model pipeline (Google Gemini 2.5 Flash) then distils negative and mixed reviews into a small set of concrete, evidence-grounded, department-owned operational issues while preserving the guest's exact specifics such as room number, amount and time. Across an analysed corpus of 2,939 reviews (weighted toward the reputation-relevant negative and mixed reviews), 2,294 problem reviews collapsed into 117 active issues, roughly a 20:1 reduction, alongside 587 single-mention early-warning signals, at a marginal detection cost of about US$0.25 per full run. The report also addresses the commercial dimensions of the system. These cover the authenticity of review data, the hotel's true booking and review volumes, a defensible and bounded estimate of the room revenue exposed to online reputation, and a concrete pricing model. We conclude that even on conservative assumptions the platform protects a multiple of its cost, and we outline the path to production through incremental detection and property-management-system integration.

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1 Introduction

1.1 Background

Travellers no longer scroll through thousands of reviews before choosing a hotel. They ask a search engine or an AI assistant a direct question, such as *"Is The Kingsbury Colombo a good hotel?"*, and receive a synthesised answer drawn from exactly those reviews. That answer typically carries a caveat along the lines of *"well-regarded, though some guests mention dated rooms, noise from live music, and occasional service inconsistency."* The caveat is not an opinion. It is a problem that surfaced in reviews, was repeated by enough guests, and became baked into how the property is described on every subsequent query. Online reputation has therefore become a direct commercial asset. Cornell's analysis of lodging performance found that a one-percent improvement in a hotel's online reputation score is associated with a 1.42% increase in revenue per available room (RevPAR) (Anderson, 2012), and more than 80% of travellers read reviews before booking (Tripadvisor, 2019; ReputationDefender, 2023).

1.2 Motivation

The mechanism that now judges the hotel, an AI reading its reviews, is precisely the mechanism a hotel should run *for itself*, but sharper and earlier. A generic assistant only reports that an issue has *already* bubbled up, and by then it is a public caveat that no tool can delete. The opportunity is to detect each operational fault at its *first* mention rather than its two-hundredth, route it to the responsible department, and fix the cause before it spreads. This is not review deletion or damage repair. It is a standing investment in the reputation the hotel has not yet lost.

1.3 Problem statement

A high-volume property generates more reviews, across more platforms and languages, than any operations team can read, de-duplicate, risk-rank and assign consistently. The cost is not the reading time. It is the *pattern that gets missed*, the single complaint today that is silently the two-hundredth of a recurring fault. The core problem this project addresses is the **cognitive-load and synthesis gap**, namely converting a continuous, multi-platform stream of unstructured reviews into a short, prioritised, evidence-backed and department-owned list of actionable issues.

1.4 Objectives

- **O1.** To **ingest and normalise** guest reviews from multiple platforms (Google, TripAdvisor, Booking.com) into a single canonical store through modular, official-shaped connectors.
- **O2.** To **analyse** each review for sentiment, owning department, and a transparent 0–100 reputation-risk score.
- **O3.** To **detect operational issues** by distilling negative and mixed reviews into a small set of concrete, de-duplicated, evidence-grounded issues that preserve the guest's specifics.
- **O4.** To **prioritise and assign** each issue to a department with risk-based ranking, recurrence tracking, and human-in-the-loop resolution.
- **O5.** To **evaluate the commercial case**, covering data validity, demand context, a bounded revenue impact, and a concrete pricing model, against a real five-star property.

1.5 Scope

The deliverable is a functioning proof-of-concept (POC) running locally, comprising a web application, an API, and a relational database. It uses official-shaped connectors with sample and crawled data rather than live production credentials, and is sized for a single property. Production concerns such as live platform authorisation, incremental detection at full scale, and property-management integration are designed for but deliberately out of POC scope, and are discussed in Sections 6 and 7.

2 Related Work

Review mining and aspect-based sentiment analysis (ABSA). Extracting opinions about specific aspects of a product or service from free text is a well-studied task, and the SemEval ABSA shared tasks established standard formulations for aspect-term extraction and polarity classification (Pontiki et al., 2016). Classical ABSA, however, produces *aspect categories* such as "rooms" or "staff" rather than the concrete, owner-assignable *incidents* an operations team can act on, and it rarely preserves the specifics, such as a room number or a billed amount, that make a complaint a closeable work order.

Online reputation and revenue. The link between online reputation and hotel performance is empirically established. Anderson (2012) quantified the RevPAR elasticity used throughout this report, and subsequent industry studies confirm that the large majority of travellers consult reviews before booking on *any* channel (Tripadvisor, 2019; ReputationDefender, 2023).

Commercial reputation and social-listening tools. Existing platforms aggregate reviews and present sentiment dashboards and word clouds. Their gap, for an operator, is threefold. They report generic themes such as "people mention food" rather than specific issues such as "breakfast buffet ran empty by 9 am". They attach no *ownership* or tracking. And they react to themes only once those themes are already prominent. This project's contribution is the combination of hotel-specific issue detection, evidence and specifics down to the room number, department ownership and tracking, and early-warning detection of single-mention signals, powered by a modern large language model (Google DeepMind, 2025) over locally-computed sentence embeddings (Reimers and Gurevych, 2019).

3 System Design and Methodology

3.1 Architecture overview

The system is a three-tier application (Figure 1). A **Next.js / React** web client presents three operator screens, namely Dashboard, Reviews and Issues. A **FastAPI** service exposes a REST API and houses the ingestion connectors, the natural-language analysis, and the issue-detection pipeline. State is persisted in **PostgreSQL**. Heavy natural-language models run locally, covering sentiment, zero-shot department classification, and sentence embeddings. Only the issue-detection reasoning calls an external large language model (Google Gemini 2.5 Flash), with a deterministic offline *stub* provider available for development and testing.

3.2 Ingestion and the canonical data model

Each platform has a dedicated **connector** that accepts the platform's own ("official-shaped") payload and maps it to a canonical review. No scraping is required, and a source can be swapped without touching the analysis engine. Raw payloads are preserved verbatim, with a content hash for idempotent and duplicate-safe ingestion, and then normalised. The data model (Figure 2) is a linear enrichment pipeline. A raw review becomes a normalised review, which gains a one-to-one analysis record, and may be linked as evidence to one or more detected issues.

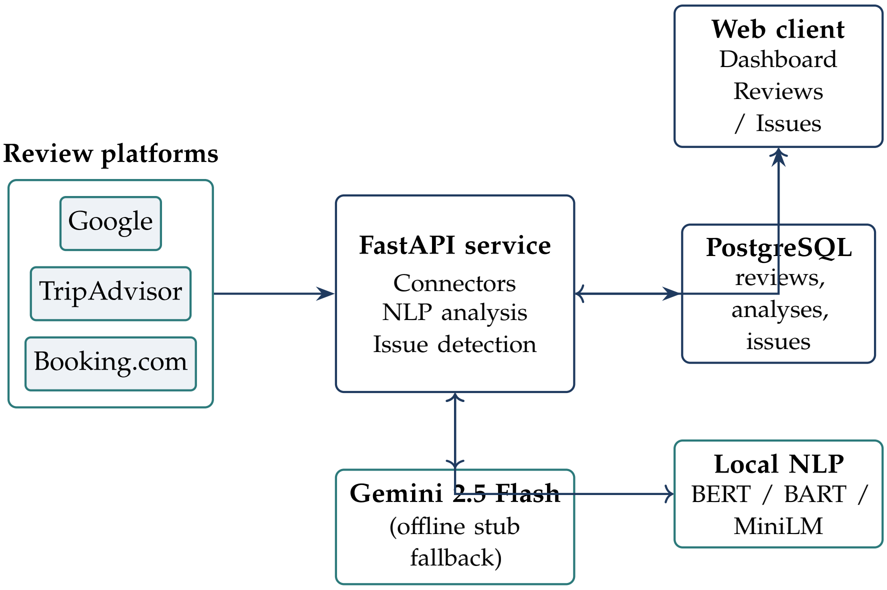

Figure 1: System architecture. Reviews flow from modular platform connectors into the FastAPI service, which analyses them with local NLP models and an external LLM, persists results in PostgreSQL, and serves the operator-facing web client.

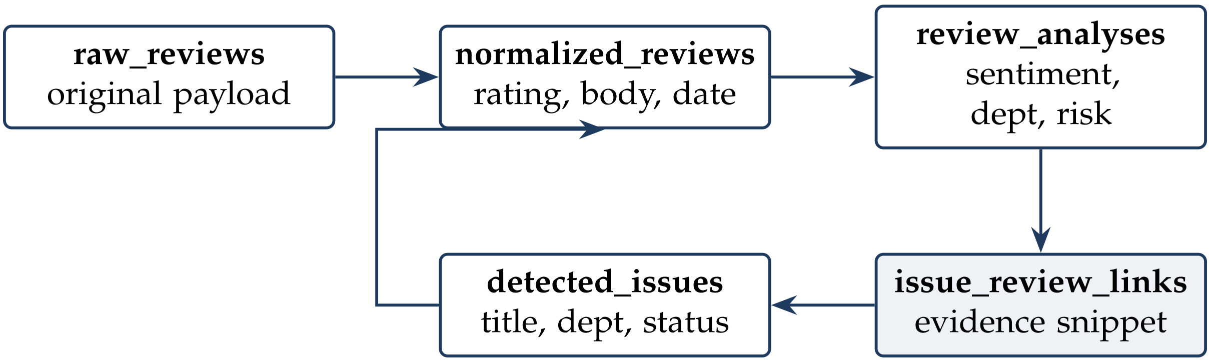

Figure 2: Canonical data model. Reviews are progressively enriched, and issues are linked back to the exact reviews and snippets that evidence them, giving every issue a traceable basis.

3.3 Natural-language analysis

Every normalised review is scored by three local models. The first is a multilingual BERT sentiment classifier (positive, mixed or negative). The second is a zero-shot department classifier built on BART-MNLI (Lewis et al., 2020) over six hotel departments (Front Office, Housekeeping, Food & Beverage, Engineering, Guest Relations, Management). The third is a MiniLM sentence-embedding model (Reimers and Gurevych, 2019) that produces a 384-dimensional vector used later for de-duplication and issue clustering.

Reputation-risk score. Each review receives a transparent, auditable 0–100 reputation-risk score, labelled low, medium, high or critical, computed as a bounded sum of interpretable factors, so a manager can see *why* a review is urgent.

$$\text{risk} = \min\Big(100,\ \underbrace{\tfrac{5-r}{4}\cdot 30}_{\text{rating}} + \underbrace{(-s)\cdot 25}_{\text{sentiment}} + \underbrace{w_d}_{\text{dept.}} + \underbrace{u}_{\text{urgency}} + \underbrace{v}_{\text{visibility}} + \underbrace{\rho}_{\text{recency}} + \underbrace{\delta}_{\text{duplicate}}\Big) \tag{1}$$

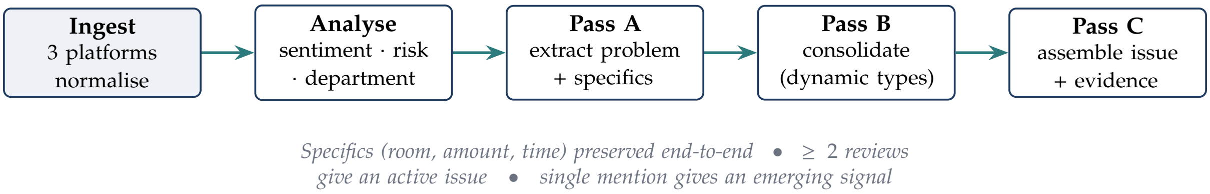

Figure 3: The end-to-end pipeline. Reviews are ingested and analysed, then a three-pass LLM stage extracts, consolidates and assembles concrete, evidence-grounded issues.

Here r is the star rating, $s \in [-1, 1]$ the sentiment score, and w_d a per-department weight (Management highest). The term u adds points for urgency words such as "refund", "unsafe" or "manager", v for public visibility through helpful votes, a public URL or unanswered status, ρ for recency, and δ for a detected duplicate. Every contributing factor is stored alongside the score for audit.

3.4 Issue detection through a three-pass LLM pipeline

The core contribution is the issue-detection pipeline (Figure 3), which runs only over **negative and mixed** reviews, the reputation-relevant ones, in three passes.

Pass A, Extract.
Reviews are sent to the LLM in small batches, and for each it returns a short problem summary, the owning department, and the concrete *specifics* such as room or floor, item, amount and time.

Pass B, Consolidate.
All extracted problem labels are clustered into a *dynamic* taxonomy of canonical issue types rather than a hard-coded list. Genuinely distinct severe incidents such as a pest sighting, theft or a safety hazard are deliberately kept separate rather than folded into a generic category.

Pass C, Assemble.
Each canonical type becomes one issue with an evidence-grounded description. A type supported by ≥ 2 reviews becomes an **active** issue, while a single-mention type becomes an **emerging** early-warning signal. A stable `cluster_key` lets the pipeline be re-run idempotently while preserving manual state such as assignee and resolved status.

The defining property is **specificity preservation**. The output is not "billing problems" but "room 1413, US$60 unexplained charge", a closeable work order rather than a theme.

3.5 Data and evaluation methodology

The analysed corpus comprises **2,939 reviews**. These combine *real reviews crawled from Google and TripAdvisor* for The Kingsbury with a set of *engineered incident fixtures*. The corpus was deliberately weighted toward **negative and mixed** reviews because those are the only reputation-actionable inputs. The detection pipeline reads negative and mixed reviews exclusively, and positive reviews yield no issues to act on. The engineered fixtures were generated from a library of specific incident scenarios, each carrying concrete facts such as room, floor, item, amount and time, precisely in order to **stress-test specificity preservation**. Real crawled reviews frequently omit such details, so synthetic incidents let us verify that the pipeline carries a guest's exact facts all the way through to the final issue. This is the principal validity caveat of the quantitative findings and is discussed further in Section 6. The macro reputation picture, namely the rating, ranking and dominant complaint themes, is corroborated by the real crawled reviews and by independent public data.

[Screenshot placeholder]

`figures/dashboard.png`

Dashboard showing the KPI cards (sentiment mix, risk distribution, and active, high-risk and recurred issue counts) with sentiment and risk donuts and per-department issue bars.

Figure 4: Dashboard showing the KPI cards (sentiment mix, risk distribution, and active, high-risk and recurred issue counts) with sentiment and risk donuts and per-department issue bars.

4 Results and Findings

4.1 Scale and compression

Table 1 summarises the headline result. The 2,294 negative and mixed reviews collapsed into 117 active issues, a roughly **20:1 compression**, so that managers act on 117 things rather than reading thousands of complaints. A further 587 single-mention *emerging* signals were caught as early warnings, and 77% of complaint reviews were traced to a concrete, fixable issue via 3,793 evidence links. The dashboard and issues screens (Figures 4 and 5) surface these for the operations team.

Table 1: Corpus and detection outcomes.

Metric	**Value**
Reviews analysed (negative/mixed-weighted corpus)	2,939
Negative & mixed (reputation-relevant) reviews	2,294
Active issues detected	117
Compression ratio (problem reviews to active issues)	≈ 20:1
Emerging (single-mention) early-warning signals	587
Complaint reviews traced to a concrete issue	77% (3,793 links)
Issues at critical risk (≥ 80)	54
Marginal LLM cost per full detection run	≈ US$0.25

4.2 The issues that matter

Table 2 lists the highest-impact detected issues. The single most-complained-about problem, loud music and event noise with 201 mentions at risk 89, is *exactly* the caveat a conversational AI repeats about the hotel, which confirms that the system measures the same signal the market already reads. Critically, issues arrive with receipt-level specificity. Billing complaints are not a generic category but, for example, "room 1413, US$60 unexplained charge", a work order the front office can close immediately.

Active issues distribute across owners as Food & Beverage 37, Front Office 25, Engineering 25, Housekeeping 15, Management 9 and Guest Relations 6, for 117 in total, giving each department head a self-contained, ranked queue.

5 Business and Market Analysis

This section examines the commercial case for the system, covering review authenticity, the hotel's true demand and review volumes, a defensible revenue impact, and pricing.

Table 2: Top detected issues (verbatim from the system).

Issue	Mentions	Risk	Owner
Loud music / event noise disturbance	201	89	Management
Air conditioning fails to cool rooms	174	74	Engineering
Long wait at check-in / check-out	169	85	Front Office
Breakfast buffet not replenished	136	72	Food & Beverage
Incorrect / unexplained charges on bill	122	82	Front Office
Slow / unstable WiFi	114	85	Engineering
Poor food quality	113	85	Food & Beverage

[Screenshot placeholder]

`figures/issues.png`

Issues page showing the active-issue queue with title, owning department, status, priority, reputation-risk score and assignee. An "Emerging candidates" tab holds the 587 single-mention signals.

Figure 5: Issues page showing the active-issue queue with title, owning department, status, priority, reputation-risk score and assignee. An "Emerging candidates" tab holds the 587 single-mention signals.

5.1 Review authenticity and data validity

A fair objection is that anyone can post a review claiming a specific room had a fault, and that the hotel's own property-management system (PMS) is the authoritative record of who stayed where. We make three points. First, two of the three sources are **verified-stay channels**. On Booking.com and, predominantly, TripAdvisor, reviews originate from guests with a booking, which constrains fabrication, whereas Google is the open channel and is treated as lower-confidence. Second, and more fundamentally, **authenticity is not the problem this system solves**. Its purpose is to reduce the cognitive load of monitoring a high-volume, multi-platform review stream and to surface recurring, reputation-damaging patterns early. A pattern of 201 independent guests reporting noise is meaningful *regardless* of any single review's provenance, because the signal is in the aggregation rather than the individual post. Third, the system never acts autonomously. Every issue carries its evidence and a confidence score, and a human resolves or dismisses it. *The system proposes, and staff decide.* Integrating the PMS, so as to corroborate a complaint against the actual folio for room 1413 on a given night and turn a flagged issue into a verified one, is a high-value enhancement, but it requires hotel-internal access and is therefore planned as production work rather than POC scope (Section 7).

5.2 Demand and review context, would the hotel care?

To judge materiality we size the hotel's demand using its real room count of 229 and clearly-flagged illustrative operating assumptions (Table 3). At a representative 72% occupancy and a US$130 average daily rate, the property turns roughly 60,000 occupied room-nights and about US$7.8 M of room revenue per year, and at an estimated average stay of about 2.5 nights it sees on the order of 65 arrivals per day. Against its roughly 20,000 cumulative public reviews, current review velocity is on the order of 10 to 20 new reviews per day across platforms. These are not large numbers to *read*, which is the heart of a natural objection, that a staff member could simply read them. But reading is not the task. The task is *synthesis*, namely connecting one complaint today to the two hundred spread across twelve months, three platforms and several languages, scoring each consistently, and spotting the one emerging issue among hundreds. That is what no human team sustains, and it is what the 20:1 compression and 587 early-warning signals deliver.

Table 3: Demand context for The Kingsbury. The room count of 229 is real, and operating ratios are illustrative benchmarks, flagged as such.

Quantity	Estimate
Rooms (actual)	229
Assumed occupancy (illustrative)	72%
Assumed average daily rate (illustrative)	US$130
Occupied room-nights / year	≈ 60,200
Room revenue / year (illustrative)	≈ US$7.8 M
Arrivals / day (at ~2.5-night stay)	≈ 65
Cumulative public reviews (all platforms)	≈ 20,000
Estimated new reviews / day	≈ 10–20

5.3 Estimating the revenue impact

A naive estimate applies the Cornell elasticity, where +1% reputation gives +1.42% RevPAR (Anderson, 2012), to the *entire* US$7.8 M room revenue, giving about US$111k per 1% of reputation protected. That figure overstates the effect, because not all revenue responds to online reputation. A portion of bookings comes through offline channels such as corporate accounts, travel agents and walk-in. We therefore estimate the impact against only the revenue base that online reputation plausibly influences, and we present a **range** rather than a single number (Table 4).

The bounds are reasoned as follows. The *floor* attributes impact only to OTA-booked room-nights. In Asia, OTAs account for roughly 70% of *online* hotel bookings (Digital Travel APAC / Cambodge Mag, 2024), which for a five-star city hotel with substantial direct and corporate business we take as about 25% of *total* room-nights. The *ceiling* recognises that review influence is not confined to OTA bookings, because more than 80% of travellers read reviews before booking on *any* channel, including the hotel's own website (Tripadvisor, 2019; ReputationDefender, 2023), so reputation shapes demand far beyond the OTA slice. The takeaway is not a single dramatic number but a robust one. **Each 1% of reputation protected is worth roughly US$28k to US$89k per year**, with a midpoint near US$60k, and the elasticity cuts both ways, so this is also the revenue *at risk* from a 1% erosion.

Table 4: Bounded annual value of protecting 1% of online reputation. The Cornell elasticity is applied to the review-influenced revenue base, and all hotel financials are illustrative.

Scenario	Influenced base	Revenue base	Value / 1%
Floor (OTA room-nights only)	≈25%	≈US$2.0 M	≈US$28k
Mid (all online bookings)	≈55%	≈US$4.3 M	≈US$61k
Ceiling (all review-influenced demand)	≈80%	≈US$6.3 M	≈US$89k

5.4 Cost analysis and pricing

The platform is inexpensive to run. The dominant marginal cost is the LLM detection run at about US$0.25 over roughly 1,000 reviews, and even run daily that is under US$100 per year.

Local NLP models run on commodity hardware at no per-use cost, and infrastructure for a single property is modest. Table 5 sets out the proposed commercial model.

Recommended model, a one-time licence plus support. The hotel pays a one-time product and implementation licence, a small monthly support fee, and the very low runtime cost directly. This suits a single flagship property that prefers a capital purchase over an open-ended subscription, and it keeps the vendor's incentives aligned with reliability rather than usage. *An alternative we considered is an annual SaaS fee*, a flat per-property annual charge with runtime folded in, which is simpler to budget and better for a multi-property rollout, and it is listed here as the secondary option.

Table 5: Proposed pricing under the recommended model, with indicative figures.

Component	Indicative
One-time product licence + implementation	≈US$9,500
Monthly support & maintenance	≈US$200 / month
LLM runtime (paid by hotel, daily runs)	<US$100 / year
Year-1 total (illustrative)	≈US$12,000
Ongoing annual (support + runtime)	≈US$2,500

Return on investment. Even on the conservative floor of Table 4, protecting just half a percent of reputation is worth about US$14k per year, which is above the year-one cost, and on the midpoint the platform pays for itself many times over. Because the value scales with revenue protected while the cost is near-fixed, the return is order-of-magnitude rather than marginal.

6 Discussion, Limitations, Risks and Mitigations

Synthetic-data validity. The clearest limitation is that the quantitative corpus is augmented with engineered incident fixtures (Section 3.5), so the absolute issue counts should be read as a demonstration of the pipeline's behaviour at scale rather than as a precise census of The Kingsbury's faults. The augmentation was a deliberate methodological choice to test specificity preservation, and the dominant themes are corroborated by real crawled reviews. A full production run over the hotel's complete real review history is the natural next step.

Other risks and their mitigations are summarised in Table 6.

7 Conclusion and Future Work

This report presented a working proof-of-concept that turns a high-volume, multi-platform stream of guest reviews into a short, prioritised, evidence-grounded and department-owned list of operational issues. On an analysed corpus of 2,939 reviews the system compressed 2,294 problem reviews into 117 active issues, a roughly 20:1 reduction, plus 587 early-warning signals, at a marginal cost of about US$0.25 per run, while preserving the guest's exact specifics down to room numbers and billed amounts. We grounded the commercial case in bounded, sourced figures. These covered verified-stay data with human-in-the-loop validation, a realistic demand and review-velocity picture, a revenue impact of roughly US$28k to US$89k per 1% of reputation protected against a year-one cost near US$12k, and a concrete one-time-licence-plus-support pricing model.

Table 6: Key risks and mitigations.

Risk	Mitigation
Data-source dependency	Official-shaped, modular connectors with no scraping. A source can be swapped without touching the detection engine.
LLM cost at scale	The pipeline already stores a cluster centroid per issue, which enables *incremental* detection. New reviews are matched to existing clusters cheaply and the LLM is called only for genuinely novel problems, so cost tracks new volume rather than total history.
Detection accuracy and false positives	Every issue is evidence-linked and confidence-scored with human-in-the-loop resolve or dismiss. The system proposes, and staff decide.
Adoption ("another dashboard")	The output is a prioritised, department-owned action list that fits existing operations rather than a raw feed, with near-zero cost to trial.

The principal next steps are a full detection run over the hotel's complete real review history, the incremental-detection path that bounds LLM cost at production scale, PMS integration to corroborate complaints against actual folios and convert flagged issues into verified ones, and multi-property support to amortise the platform across a hotel group. The core thesis stands. A hotel cannot delete what an algorithm has already learned about its past, but for cents per analysis it can ensure tomorrow's problems never become the next permanent caveat.

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